



Aryabrata Basu

Curriculum Vitae

*"Science is built up of facts, as a house is built of stones;
but an accumulation of facts is no more a science than a
heap of stones is a house." – Henri Poincaré, Science and
Hypothesis, 1905*

Current Positions

- 2022–present **Assistant Professor (Tenure Track)**, *Department of Computer Science, University of Arkansas at Little Rock, Little Rock, AR, United States*
- 2022–present **Research Fellow**, *Donaghey Emerging Analytics Center, University of Arkansas at Little Rock, Little Rock, AR, United States*
- 2025–present **Undergraduate Program Coordinator**, *Department of Computer Science, University of Arkansas at Little Rock, Little Rock, AR, United States*

Past Positions

- 2016–2022 **Visual Information Specialist**, *Emory Center for Digital Scholarship, Emory University, Atlanta, GA, United States*
- 2010–2016 **Graduate Research Assistant**, *University of Georgia, Athens, GA, United States*
- 2009–2010 **Graduate Teaching Assistant**, *University of Georgia, Athens, GA, United States*

Education

- 2009–2018 **Ph.D. - Computer Science**, *University of Georgia, Athens, GA, United States*
- 2003–2007 **Bachelor of Technology - Information Technology**, *West Bengal University of Technology, Kolkata, West Bengal, India*

PhD Dissertation

- Dissertation [Designing and evaluating ubiquitous wearable virtual reality](#)
- Advisor(s) Johnsen K, Potter D, Rasheed K, and Liu T

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Research Interests

Topics Human-Computer Interaction, Virtual Reality + Augmented Reality = Extended Reality, Virtual Worlds, Virtual Avatars, Animation, Natural User Interfaces (2D/3D), Usability Analytics, Study Design, Computer Graphics, Artificial Intelligence, Machine Learning, Distributed Computing, Robotics, Prototyping wearable hardware and its design, and Haptics.

Honors and Recognition

- 2025 **Best Paper Award ('Honorable Mention')** - 'From Novelty to Knowledge: A Longitudinal Investigation of the Novelty Effect on Learning Outcomes in Virtual Reality', IEEE Virtual Reality.
- 2014 **Best Paper Award** - 'Mixed Reality Virtual Pets to Reduce Childhood Obesity', IEEE Virtual Reality.
- 2013 **Best Poster Award** - 'Evolution and Usability of Ubiquitous Immersive 3D Interfaces', IEEE Symposium on 3D User Interfaces.
- 2010 **Best Poster Award ('Honorable Mention')** - 'Field Valid Probability Assessment in Uncertain and Multi-agent Settings', Research Day 2010, Department of Computer Science, University of Georgia.

Media and Public Engagement

- 2025 **America250 – [Our American Story](#)**, Featured Interviewee (sole researcher profile from the State of Arkansas). Library of Congress archival project; filmed at Old State House, Little Rock, AR. Oct 2025.
- 2025 **KUAR 89.1 (NPR member station), Morning Edition** — Guest interview on [Mapping cardiac blood vessels in 3D](#). Air date: Sep 11, 2025.

Grants and Sponsored Research Activity

Funded Awards

- 2024–2025 **Cybersecurity Consortium for Innovation** - U.S. Department of Energy, National Energy Technology Laboratory, \$5.0 million total funding. Role: Co-Investigator. Focus: immersive industrial control systems cybersecurity training, workforce development, and cyber-physical simulation.
- 2023–2024 **Robust 2D to 3D Cardiac Arterial Mapping and Visualization for Surgical Preplanning** - NSF/EPSCoR-DART, \$100,000 total funding. Role: Principal Investigator. Focus: digital cardiology, conformal arterial mapping, 3D visualization, and surgical preplanning.
- 2023–2024 **2D to 3D Cardiac Arterial Mapping and Visualization** - EPSCoR-DART, \$66,000 total funding. Role: Principal Investigator. Focus: cardiac arterial reconstruction, immersive visualization, and medical decision support.

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Active Proposal Portfolio

Advanced Air Mobility and Digital Twins	NASA EPSCoR FY 2026 Basic Research: Human-in-the-Loop Digital Twin for Noise-Aware Regional AAM Fleet Planning (pending). Role: Principal Investigator. Focus: human-in-the-loop digital twins, Advanced Air Mobility/UAS corridor planning, noise-aware fleet scenario comparison, acoustic exposure and human-response modeling, telemetry replay, 3D geospatial visualization, and NASA-aligned regional aviation decision support.
Quantum and Nanotechnology Infrastructure	NSF NQNI: Arkansas Open-Access Quantum and Nanotechnology User Facility for Secure User Enablement, Digital Twin Workflows, and Workforce Development (pending). Role: Co-Principal Investigator. Focus: open-access nanotechnology and QISE user infrastructure, secure immersive user enablement, digital-twin onboarding, AI-enabled discovery workflows, workforce training, external-user support, and Mid-South regional research capacity.
Immersive AI and Human-Centered Computing	NSF CAREER: Persistent Virtual Reality Spaces; Secure Analytics for Immersive Environments (in preparation). Focus: persistent immersive environments, embodied interaction, privacy-preserving behavioral telemetry, responsible immersive analytics, human-centered AI, and AI-mediated learning.
Secure XR Training and Evidence-Based Assessment	ORETTC / Y-12: SPECTRA - Secure Platform for Encrypted Content Transmission, Replay, and Analytics (pending). Role: Principal Investigator. Focus: secure real-time 3D mesh distribution, encrypted XR content streaming, distributed digital twin training, Unity-based immersive simulation, ELK telemetry ingestion, behavioral replay, usability analytics, after-action review, and evidence-based training assessment for national security, nuclear operations, cybersecurity, and emergency preparedness contexts.
Scientific and Medical Visualization	Digital Cardiology; Open Carbon Atlas; Remote Sensing-Assist Crop Phenology Detection and Immersive Visualization (in preparation). Focus: AI-assisted scientific infrastructure, 3D medical visualization, carbon science, agricultural decision support, remote sensing, and immersive data interaction.
AI Assurance, Sensing, and Cyber-Physical Awareness	DARPA Young Faculty Award; AFOSR/ARO Conformal Augmented Reality for Passive Wireless Signal Decoding (in preparation). Focus: formal assurance, loss-of-control containment of AI, passive RF sensing, electromagnetic awareness, conformal mapping, and augmented reality visualization.
Society, Law, and Scientific Communication	NSF Science of Science: Discovery, Communication, and Impact; NSF Law and Science: Spatially Queryable Crime Scene Simulation Platform (in preparation). Focus: computational analysis of scientific discourse, scientific correspondence, communication media, forensic simulation, spatial querying, and law-science research infrastructure.
Edge Intelligence and Immersive Infrastructure Digital Twins	NSF VINES Track 1: Augmented Reality and Edge-Aware Digital Twins for Multi-Utility Infrastructure Management (in preparation). Role: Co-Principal Investigator / Senior Personnel. Focus: edge-resident digital twins, multi-utility infrastructure modeling, Bayesian filtering, distributed and federated learning, resource-aware 3D visualization, augmented reality field interfaces, privacy-preserving telemetry, and human-centered evaluation for utility maintenance.

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Teaching

- 2025 **Professor**, *Capstone Projects (CPSI 49203), Adv Game Programming (CPSI 37703), Seminar in Computer Architecture (CPSI 73103)*, UA Little Rock, Arkansas
Academic Year
- 2024 **Professor**, *Computer Networks (CPSC 3384), Database Concepts (CPSC 3375), Computer Architecture (CPSC 7331), Distributed Computing (CPSC 43/5387)*, UA Little Rock, Arkansas
Academic Year
- 2023 **Professor**, *Computer Networks (CPSC 3384), Database Concepts (CPSC 3375), Computer Architecture (CPSC 7331), Distributed Computing (CPSC 43/5387)*, UA Little Rock, Arkansas
Academic Year
- Fall 2022 **Professor**, *Distributed Computing (CPSC 4387/5387)*, UA Little Rock, Arkansas
- 2019 – 2022 **Lead Instructor**, *Introduction to 3D Visualization and Interactive Media Design (ARTHIST 393R)*, Emory University, Atlanta

Service

- 2025-26 **Program Committee Member**: IEEE International Conference on Artificial Intelligence & eXtended and Virtual Reality (AIxVR 2026)
- 2025 **Editorial Board Member**: [Nature Portfolio](#)
- 2025 **Track Chair(Main+Posters)**: The ACM Symposium on Applied Perception (SAP)
- 2025 **Program Committee Member**: TRUST-XR 2025 @IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2024-25 **Program Committee Member**: IEEE International Conference on Artificial Intelligence & eXtended and Virtual Reality (AIxVR 2025)
- 2024 **Panel Reviewer**: [NEH](#)
- 2021 **University Senate Committee Member**: [COE](#)
- 2020 **Panel Reviewer**: [NSF](#)
- 2020 **Associate Editor**: [Presence](#)
- 2020 **Review Editor**: [Frontiers in Virtual Reality](#)
- 2020 **Review Editor**: [Springer Nature Virtual Reality](#)
- 2020-21 **Program Committee Member**: International Symposium on Visual Computing
- 2020 **Session Chair**: Immersive Learning Research Network
- 2020 **Local Arrangements Chair, Usability Study Researcher**: IEEE Virtual Reality
- 2012-2013 **Student Volunteer**: IEEE Virtual Reality
- 2012 **Student Volunteer**: IEEE International Symposium on Mixed and Augmented Reality

Membership

- IEEE Senior Member
- ACM Lifetime Member

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Publications

- Sunny, M. J. M., Springer, J. P., and Basu, A. Evaluating adjustment and proficiency disparities in virtual reality. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pages 1282–1283. IEEE, 2025.
- Sunny, M. J. M., Guthula, J. S. N., Kharel, A., Shraboni, M. N., Rashid, H., Bhandari, P., Springer, J. P., and Basu, A. Gaze insights in xr: Real-time eye-tracking analytics with elasticsearch. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pages 97–103. IEEE, 2025.
- Lee, J., Chen, C. C., and Basu, A. From novelty to knowledge: A longitudinal investigation of the novelty effect on learning outcomes in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 31(5):3204–3212, 2025.
- Guthula, J. S. N., Rashid, H., Springer, J. P., and Basu, A. Preserving privacy in vr telemetry data. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pages 1270–1271. IEEE, 2025.
- Bhandari, P., Young, C. Z., Springer, J. P., and Basu, A. Scientific crosstalk: Analyzing sentiment-topic alignment in personal and scientific discourse. In *International Conference on Machine Learning and Applications*, pages 99–106. IEEE, 2025.
- Tullis, J. A., Lanter, D. P., Basu, A., Cothren, J. D., Shi, X., Limp, W. F., Linck, R. F., Young, S. G., Davis, J., and Alsumaiti, T. S. Geoprocessing, workflows, and provenance. In *Remote Sensing Handbook, Volume II*, volume 2, pages 303–337. CRC Press, 2024.
- Sunny, M. J. M. and Basu, A. Non-linear parameterization of spatial decision making in immersive virtual environment. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pages 389–395. IEEE, 2024.
- Guthula, J. S. N. and Basu, A. Navigating gender biases in xr: Towards equitable technological future. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pages 326–329. IEEE, 2024.
- Page, M., Kang, Y., Varner, S., Lee, K., Cors, A., Basu, A., Durante, K., and Burr, I. Methods and evaluation in the historical mapping of cities. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-4/W7-2023:155–160, 2023.
- Basu, A., Sunny, M. J. M., and Guthula, J. S. N. Privacy concerns from variances in spatial navigability in vr. *arXiv preprint arXiv:2302.02525*, 2023.
- Basu, A. Stag: A tool for realtime replay and analysis of spatial trajectory and gaze information captured in immersive environments. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pages 43–45. IEEE, 2022.
- Martin, A. J., Stearns, D., Whitten, M. J., Hage, M. M., Page, M., and Basu, A. First known trace fossil of a nesting iguana (pleistocene), the bahamas. *PLOS ONE*, 15(12):e0242935, 2020.
- Basu, A. Work-in-progress—augmented reality at scale: Using physical dimensionality of spaces to our advantage! In *6th International Conference of the Immersive Learning Research Network*, pages 373–376. IEEE, 2020.
- Basu, A. Work-in-progress—tracking the untracked. *arXiv preprint arXiv:1909.05327*, 2019.
- Basu, A. Work-in-progress—a brief chronology of virtual reality. *arXiv preprint arXiv:1911.09605*, 2019.

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- Basu, A. and Johnsen, K. Work-in-progress—navigating a maze differently: A user study. *arXiv preprint arXiv:1805.09454*, 2018.
- Basu, A., Ball, C., Manning, B., and Johnsen, K. Effects of user physical fitness on performance in virtual reality. In *IEEE Symposium on 3D User Interfaces*, pages 233–234. IEEE, 2016.
- Ahn, S. J. G., Johnsen, K., Robertson, T., Moore, J., Brown, S., Marable, A., and Basu, A. Using virtual pets to promote physical activity in children: An application of the youth physical activity promotion model. *Journal of Health Communication*, 20(7):807–815, 2015.
- Johnsen, K., Ahn, S. J., Moore, J., Brown, S., Robertson, T. P., Marable, A., and Basu, A. Mixed reality virtual pets to reduce childhood obesity. *IEEE Transactions on Visualization and Computer Graphics*, 20(4):523–530, 2014.
- Basu, A. and Johnsen, K. Ubiquitous virtual reality 'to-go'. In *IEEE Virtual Reality*, pages 161–162. IEEE, 2014.
- Basu, A., Johnsen, K., Bogert, K., and Wins, P. Immersive virtual reality on-the-go. In *IEEE Virtual Reality*, pages 193–194. IEEE, 2013.
- Basu, A., Johnsen, K., and Bogert, K. Poster: Evolution and usability of ubiquitous immersive 3d interfaces. In *IEEE Symposium on 3D User Interfaces*, pages 131–132. IEEE, 2013.
- Wins, P., Basu, A., and Johnsen, K. Off-the-shelf electronics prototyping for virtual reality. In *IEEE VR Workshop on Off-the-Shelf Virtual Reality*, 2012.
- Wins, P., Basu, A., and Johnsen, K. Do-it-yourself interface device prototyping for virtual reality. *International Journal of Virtual Reality*, 11(3):43–48, 2012.
- Basu, A., Saupe, C., Refour, E., Raij, A., and Johnsen, K. Immersive 3d ui on one dollar a day. In *IEEE Symposium on 3D User Interfaces*, pages 97–100. IEEE, 2012.
- Basu, A., Raij, A., and Johnsen, K. Ubiquitous collaborative activity virtual environments. In *Proceedings of the ACM 2012 Conference on Computer Supported Cooperative Work*, pages 647–650. ACM, 2012.
- Mrázek, J., Chaudhari, T., and Basu, A. Perplot & perscan: Tools for analysis of dna curvature-related periodicity in genomic nucleotide sequences. *Microbial Informatics and Experimentation*, 1(1):13, 2011.
- Basu, A. and Nachttegaal, M. Owa filters: A robust filtering method and its application to color images. In *IEEE Symposium on Computational Intelligence for Image Processing*, pages 8–13. IEEE, 2009.
- Basu, A. and Funk, S. An optimal scheme for multiprocessor task scheduling: A machine learning approach. In *Work-In-Progress Proceedings, The 30th IEEE Real-Time Systems Symposium*, page 29. Citeseer, 2009.